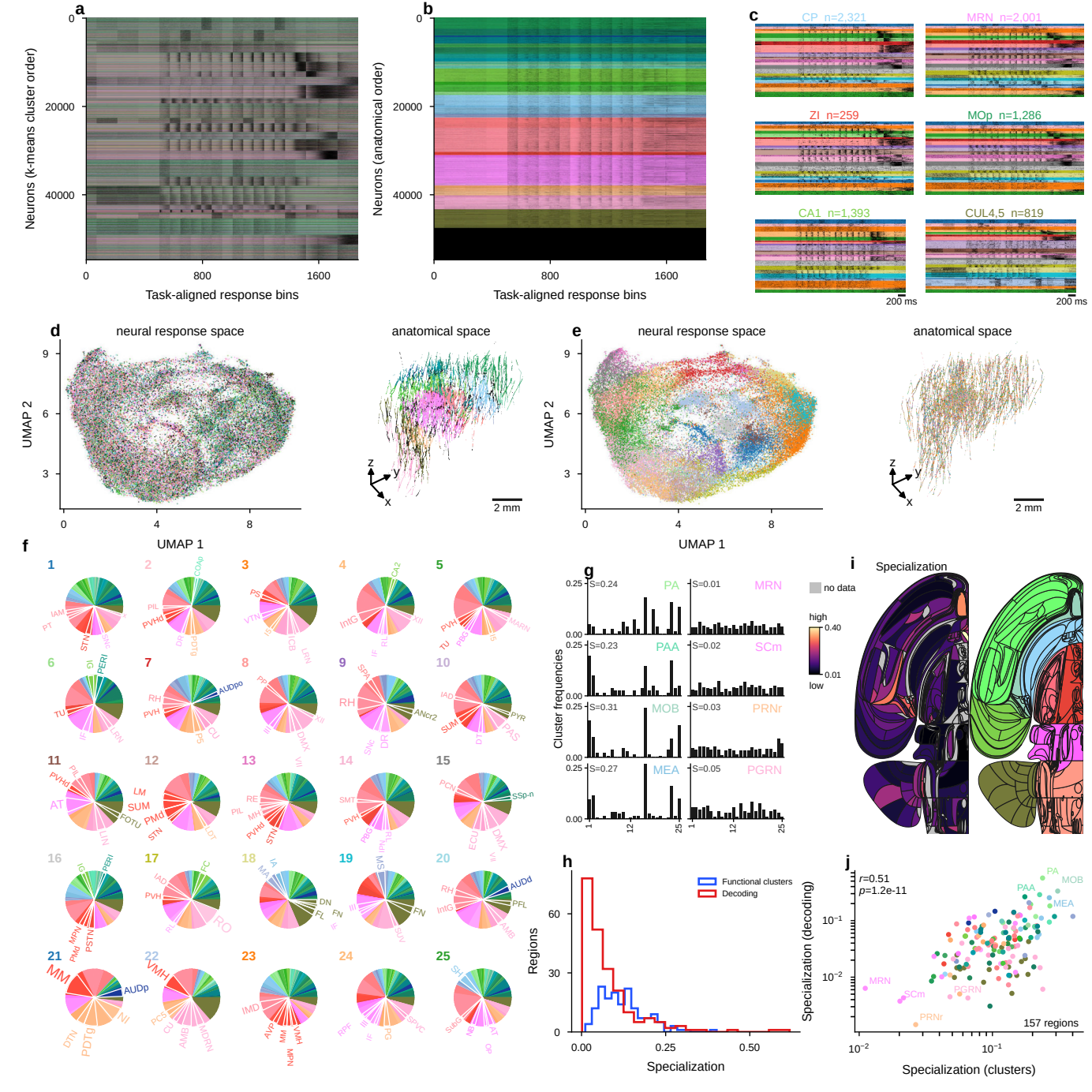

a
Neurons (k-means cluster order)
0
20000
40000
0
800
1600
Task-aligned response bins
b
Neurons (anatomical order)
0
20000
40000
0
800
1600
Task-aligned response bins
c
CP n=2,321
MRN n=2,001
ZI n=259
MOp n=1,286
CA1 n=1,393
CUL4,5 n=819
200 ms
200 ms
d
neural response space
anatomical space
UMAP 2
9
6
3
0
4
8
z
y
x
2 mm
e
neural response space
anatomical space
UMAP 2
9
6
3
0
4
8
UMAP 1
z
y
x
2 mm
f
UMAP 1
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
g
Cluster frequencies
0.25
0.00
S=0.24
PA
S=0.01
MRN
S=0.23
PAA
S=0.02
SCm
S=0.31
MOB
S=0.03
PRNr
S=0.27
MEA
S=0.05
PGRN
1
12
25
i
Specialization
no data
high
0.40
0.01
low
h
Regions
60
30
0
Functional clusters
Decoding
0.00
0.25
0.50
Specialization
j
r=0.51
p=1.2e-11
PA
MOB
PAA
MEA
MRN
SCm
PGRN
PRNr
157 regions
Specialization (decoding)
10^{-1}
10^{-2}
10^{-2}
10^{-1}
Specialization (clusters)